

| S/N | Question                                                                     | Options                                                                                                                                                                                                                                                                                                                                     | ANS |
|-----|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 5   | The “feasibility study” is performed during which SDLC phase?                | A) Planning B) Development C) Testing D) Maintenance                                                                                                                                                                                                                                                                                        | A   |
| 17  | Which of the following is equivalent to $\neg(p \vee q)$ ?                   | A) $\neg p \wedge \neg q$ B) $\neg p \vee \neg q$ C) $p \wedge q$ D) $p \vee q$                                                                                                                                                                                                                                                             | A   |
| 20  | The statement “It is not the case that both p and q are true” is written as: | A) $\neg(p \wedge q)$ B) $(\neg p) \wedge (\neg q)$ C) $p \rightarrow \neg q$ D) $p \vee \neg q$                                                                                                                                                                                                                                            | A   |
| 24  | Which of the following truth tables corresponds to $p \leftrightarrow q$ ?   | A) T,T $\rightarrow$ T F,F $\rightarrow$ T T,F $\rightarrow$ F F,T $\rightarrow$ F B) T,T $\rightarrow$ F F,F $\rightarrow$ T T,F $\rightarrow$ T F,T $\rightarrow$ F C) T,T $\rightarrow$ T F,F $\rightarrow$ F T,F $\rightarrow$ F F,T $\rightarrow$ T D) T,T $\rightarrow$ T F,F $\rightarrow$ T T,F $\rightarrow$ T F,T $\rightarrow$ F | A   |
| 27  | The <b>B-Method</b> was created by which researcher?                         | A) Jean-Raymond Abrial B) Bertrand Meyer C) Niklaus Wirth D) Robert C. Martin                                                                                                                                                                                                                                                               | A   |
| 29  | The term “ <b>model checking</b> ” was coined by _____.                      | A) Edmund M. Clarke B) Leslie Lamport C) John McCarthy D) Brian Kernighan                                                                                                                                                                                                                                                                   | A   |
| 1   | Which SDLC stage defines what the system must do?                            | A) Design B) Requirements C) Testing D) Maintenance                                                                                                                                                                                                                                                                                         | B   |
| 3   | Which stage focuses on bug detection before release?                         | A) Design B) Testing C) Requirements D) Maintenance                                                                                                                                                                                                                                                                                         | B   |
| 4   | The post-release activities (bug fixes, updates) are part of which phase?    | A) Deployment B) Maintenance C) Design D) Analysis                                                                                                                                                                                                                                                                                          | B   |
| 6   | In logic, a <b>proposition</b> is:                                           | A) A variable that can change value B) A declarative statement that is either true or false C) A program function D) A data structure                                                                                                                                                                                                       | B   |
| 8   | A proposition that is always true is called a _____.                         | A) Contradiction B) Tautology C) Paradox D) Contingency                                                                                                                                                                                                                                                                                     | B   |
| 9   | A proposition that is always false is known as a _____.                      | A) Tautology B) Contradiction C) Contingency D) Predicate                                                                                                                                                                                                                                                                                   | B   |
| 10  | The logical value of a proposition is called its _____.                      | A) Symbol B) Truth-value C) Operator D) Variable                                                                                                                                                                                                                                                                                            | B   |
| 11  | Which symbol represents logical <b>AND</b> ?                                 | A) $\vee$ B) $\wedge$ C) $\neg$ D) $\rightarrow$                                                                                                                                                                                                                                                                                            | B   |
| 12  | The operator $\neg$ stands for:                                              | A) OR B) NOT C) IMPLIES D) EQUIVALENT                                                                                                                                                                                                                                                                                                       | B   |

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| 14  | The symbol $\leftrightarrow$ is used for:                                                                             | A) Exclusive OR B) Bi-conditional (iff) C) NAND D) NOR                                     | <b>B</b> |
| 16  | Let $p$ = "It rains", $q$ = "The ground is wet". Which proposition expresses "If it rains then the ground is wet"?    | A) $p \wedge q$ B) $p \rightarrow q$ C) $q \rightarrow p$ D) $\neg p \vee q$               | <b>B</b> |
| 18  | If $p$ is true and $q$ is false, what is the truth value of $p \wedge q$ ?                                            | A) True B) False C) Undetermined D) Both                                                   | <b>B</b> |
| 19  | "Either the system is safe <b>or</b> it is insecure" can be symbolized as:                                            | A) $S \wedge I$ B) $S \vee I$ C) $\neg S \vee I$ D) $S \leftrightarrow I$                  | <b>B</b> |
| 22  | In a truth table, the column for $\neg p$ is the _____ of the column for $p$ .                                        | A) Same B) Inverse C) Duplicate D) Random                                                  | <b>B</b> |
| 23  | For the expression $p \rightarrow q$ , the row $p = \text{True}, q = \text{False}$ yields what result?                | A) True B) False C) Undefined D) Both                                                      | <b>B</b> |
| 25  | The expression $(p \vee q) \wedge \neg r$ is <b>false</b> when $p = \text{False}, q = \text{True}, r = \text{True}$ . | A) True B) False C) Cannot tell D) Both                                                    | <b>B</b> |
| 28  | The first commercial tool for formal specification was _____ (1990s).                                                 | A) SPARK B) Z/EVES C) ACL2 D) PVS                                                          | <b>B</b> |
| 2   | The coding activity belongs to which SDLC phase?                                                                      | A) Planning B) Design C) Development D) Deployment                                         | <b>C</b> |
| 7   | Which of the following is <b>not</b> a proposition?                                                                   | A) "The sky is blue." B) " $x + 2 = 5$ ." C) "Close the window." D) "7 is a prime number." | <b>C</b> |
| 13  | In propositional logic, $\rightarrow$ denotes:                                                                        | A) Conjunction B) Disjunction C) Implication D) Negation                                   | <b>C</b> |
| 21  | How many rows does the truth table for $p \wedge q \wedge r$ contain?                                                 | A) 4 B) 6 C) 8 D) 2                                                                        | <b>C</b> |
| 26  | Which famous mathematician's work on <b>predicate logic</b> laid groundwork for formal methods?                       | A) Euclid B) Alan Turing C) Gottlob Frege D) Donald Knuth                                  | <b>C</b> |
| 30  | Formal methods saw a major boost after the _____ disaster (1999) that highlighted the need for rigorous verification. | A) Therac-25 B) Mars Climate Orbiter C) Ariane 5 D) Y2K                                    | <b>C</b> |

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|-----|----------------------------------------|---------------------------------------------------|------------------------------------------------------------------------|
| 15  | Which operator is <b>commutative</b> ? | $A \rightarrow B \leftrightarrow C \neg D \wedge$ | <b>D</b> (also $\leftrightarrow$ , but $\wedge$ is the classic answer) |